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(a)(i)	The *volt* is defined as the potential difference between two points in a circuit where 1 J of electrical energy is converted to other forms of energy when 1 C of charge passes from one point to the other.
(a)(ii)	e.m.f. is the amount of non-electrical energy converted into electrical energy per unit charge passing through the terminals of the cell (source).
	p.d. is the amount of electrical energy converted to other forms of energy per unit charge passing from one point to the other.
(b)(i)	When current in P is 0.15 A, the p.d. across P is 2.70 V. The p.d. across Q is also 2.70 V. From the I-V graph of Q, the current through Q is 0.0900 A.
	Total current in battery = current in P + current in Q = 0.15 + 0.0900 = 0.24 A
(b)(ii)	P.d. across R = 4.5 – 2.70 = 1.8 V
	Resistance of R = $\frac{1.8}{0.24} = 7.5\ \Omega$
(b)(iii)	In the given circuit, Resistance of P = $\frac{2.70}{0.15} = 18\ \Omega$ Resistance of Q = $\frac{2.70}{0.0900} = 30.0\ \Omega$
	Since resistance $R = \frac{\rho L}{A}$, $L = \frac{RA}{\rho}$ Since $A \propto d^2$, $L \propto Rd^2$ $\because \rho = \text{constant}$
	$\frac{L_P}{L_Q} = \left(\frac{R_P}{R_Q}\right)\left(\frac{d_P}{d_Q}\right)^2$ $= \left(\frac{18}{30}\right)\left(\frac{2}{1}\right)^2$ $= 2.4$
(b)(iv)	When Q stops conducting, effective resistance of the two lamps increases. By potential divider principle, the p.d. across lamp P increases. The current through P also increases as no current passes through Q.
	From the graph, as p.d. across P (or current in P) increases, its resistance increases.
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